

DFG Re20 cylinder: diagnostic evidence

Physical problem: parabolic inlet, channel 2.2×0.41 m, cylinder $D=0.1$ m, center (0.2,0.2), mean speed 0.2 m/s, $\rho=1$, $\mu=0.001$, $Re=20$.

Reference: FeatFlow DFG 2D-1; $C_d=5.57953523384$, $C_l=0.010618948146$, front/rear pressure difference= 0.11752016697 Pa.

Three separate strict 3% physical gates plus residual/mass gates.

Body-fitted SIMPLE/Rhie-Chow production solver; polygon wall traction.
Pressure difference: least-squares wall extrapolation and angle interpolation.
Raw field, mesh, pressure/viscous traction, corrected wall gradients and authoritative mass flux exported. Independent NumPy integration audit.

64x24: converged=True, physical gates=False

drag: 6.32479236, error 13.3570%

lift: 0.015101714, error 42.2148%

pressure_difference_pa: 0.130991489, error 11.4630%

Single-grid diagnosis does not establish grid convergence.

No matched LBM cylinder run; no industrial or icebreaking qualification.

Next: refine geometry, transport and wall-pressure/traction reconstruction.

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